Table 1 Features of selected methods for assessing body composition in elite athletes

Method (approach)	Outcome measures	Assumptions/cautions	Accuracy	Precision/repeatability (technically)*	Advantages	Limitations
Anthropometry (skinfolds) (anatomical)	Skinfold thickness (compressed) Skinfold sum Skinfold ratios Conversion to BF (%) via equations not supported[92]	Several assumptions are required,[99] including: Constant skinfold compressibility Constant SAT compressibility Constant skin thickness Unknown proportion of embedded fibrous structures	Unknown	Intra-tester 5% TEM for ISAK-trained, skilled technicians[100]	Applicable in the field Standardised protocols Data norms available Non-invasive and no radiation Scores are minimally affected by exercise, hydration or ingestion of food Cost efficient	Measurer training necessary Samples of the subcutaneous fat deposit only Some sites difficult to achieve Measures primarily skin thickness in lean persons Can be intrusive for some individuals Some site locations not scaled to body dimensions
Ultrasound (anatomical)	SAT thicknesses (uncompressed) with fibrous structures included and excluded SAT mass SAT patterning	No image distortion with a linear probe Use the speed of sound in SAT for distance determination Use standardised protocol[101] Correct detection of tissue layer boundaries need training and experience Analysis must avoid some anatomical structures (blood vessels)	Accuracy 0.1 mm (18 MHz probe); 0.2 mm (10 MHz)	95% LOA=0.2 mm (for the mean of eight sites), which translates to 0.2 kg SAT mass[102]	Applicable in the field Standardised protocols Non-invasive and no radiation Scores are minimally affected by exercise, hydration or ingestion of food[103] High accuracy and precision of measurement of subcutaneous fat Scan site locations relative to stature No tissue compression Tissue thickness 0.1–100 mm	Measurer training necessary Samples the subcutaneous fat deposit only Expensive equipment
DXA (chemical)	Directly BMC Indirectly FM LBM	Interpolation for soft tissues in areas where bone is detected (40–45% of pixels affected)[92] Use standardised, best practice protocols [96] Awareness of misconceptions regarding which tissues comprise FM and LBM[98] Animal models used for human soft tissue calibration[98] Assumptions required and correction factors applied[104] when estimating soft tissue composition	Compared with multi-component models[105] %FM 2.0–3.0% SEE SM 1.6–4.4 kg SEE DXA calibration does not accommodate the physiques of lean athletes[101]	Scan-rescan analysis with repositioning[106] BMC 0.60% CV† FM 0.82% CV %FM 0.86% CV In ‘athletic’ populations[96] BMC 0.70% TEM FM 1.90% TEM LBM 0.40% TEM	Standardised protocols Data norms for BMD available Good precision for bone mineral Time efficient Small single scan radiation dose Whole-body approach Regional compartment analysis Provides estimate of whole body and regional FM, LBM and BMD	Not applicable to field work Inter-machine and inter-manufacturer variability Consideration of cumulative X-ray dose for multiple scans Cannot scan if pregnant Provides an indirect measure of LBM Calculation algorithms differ between manufacturers and are not published Pencil vs fan-beam outcomes differ Expensive equipment Restrictive exercise, fasting and hydration requirements prior to scanning (for athletes)

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Table 1 Continued

Method (approach)	Outcome measures	Assumptions/cautions	Accuracy	Precision/repeatability (technically)*	Advantages	Limitations
Medical imaging MRI (anatomical)	Cross-sectional areas of Adipose Bone LBM Other tissues	MRI designed primarily for diagnostic use rather than quantifying tissue masses Calculation of tissue volumes and masses requires bespoke software Image segmentation (contour detection) requires subjective decisions Relating anatomical dimensions to tissue masses requires knowledge of tissue densities	Pixel size (~1.5×1.5 mm) limits accuracy for whole body scans, especially for lean persons[92] Consequently, MRI cannot detect accurately the fat layers below this pixel size	Unknown for body fat measurements	Non-invasive and no radiation Whole-body approach possible Regional compartment analysis possible Direct measure of skeletal muscle at site of assessment	Not applicable to field work Lack of published normative data Difficulties in discriminating accurately tissue layer boundaries Bespoke software necessary for quantification of tissue masses Long data capture for whole-body scans Confined space may induce claustrophobia Single or few slices are not representative of visceral or whole-body fat mass[107] Expensive equipment
Impedance (chemical)	Total body water (TBW) Body fat (BF) Fat-free mass (FFM)	Participant compliance to strict testing prerequisites assumed – including abstaining from exercise Assumes geometric similarity between individuals Assumes tissue resistivity is similar between individuals Input data (age, height, weight, athletic status) accounts for high (up to 85%) of variance in the outcome variable.	Compared with multi-component models for young adults[108] FFM—SEE 2.5–3.9 BF—SEE 3.6 TBW—SEE 1.7–3.8	Intra-individual CV 2.0–3.5%[108] Prediction errors 3.0–8.0% for TBW 3.5–6.0% for FFM	Minimal participant involvement Non-invasive and no radiation Rapid data acquisition Precision good for TBW Apparent sophistication	Accuracy poor – most have used questionable criterion measures for comparison Lack of standardised protocols Results affected by hydration and electrolyte status No athlete-specific equations Trunk contributes only a small proportion to total impedance Different electrode placement (arm–leg, leg–leg, arm–arm) Large variability for outcome measures between devices

*Does not necessarily consider biological variation.
†CV <5% denotes good precision; TEM <1.5% denotes acceptable reliability; 95% LOA denotes reliability is better than the given value in 95% of cases.
BF (%), percent body fat; BMC, bone mineral content; BMD, bone mineral density; CV, coefficient of variation; DXA, dual-energy X-ray absorptiometry; FM, fat mass; LBM, lean body mass; LOA, limit of agreement; SAT, subcutaneous adipose tissue; SEE, SE of the estimate; TEM, technical error of measurement.